

Mental Health and Regional Characteristics

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Background

Mental health outcomes vary across U.S. counties. These differences reflect socioeconomic conditions, demographic structures, and broader regional environments.

Prior work shows that social and economic factors are associated with mental health outcomes. Studies highlight the role of income, education, and community environments (Compton & Shim, 2015), and report links between socioeconomic disadvantage and frequent mental distress (Bolduc et al., 2025), and rural/urban disparities (Negaro et al., 2023).

However, existing studies focus on limited

indicators or use older data. This project addresses this gap using PLACES and ACS data.

Motivation

Mental health is a major public-health concern. Regional characteristics can help identify community-level patterns that individual-level data may not capture. County-level analysis helps examine how community environments and socioeconomic conditions shape mental-health outcomes.

By integrating PLACES health measures with ACS socioeconomic and demographic indicators, this project provides **a more comprehensive and up-to-date assessment of county-level mental-health patterns**. The insights may support public-health planning, resource allocation, and future research on

social determinants of mental health.

Questions

This project investigates the following core question:

- **Which regional characteristics are most strongly associated with mental health across U.S. counties?**

To address this, we examine candidate factors such as:

- Income levels and poverty rates
- Employment and labor-market conditions
- Educational attainment
- Housing and household characteristics
- Other socioeconomic and demographic indicators available in the datasets

The goal is to identify regional characteristics that show the strongest associations with frequent distress across U.S. counties.

Data Sources

Primary Dataset: PLACES: County Data (GIS Friendly Format), 2025 release

County-level estimates for multiple health measures across the US, including mental health status MHLTH_AdjPrev. The estimates are calculated from BRFSS, U.S. Census, and ACS data.

MHLTH_AdjPrev represents a **model-based estimate for age-adjusted prevalence of frequent mental distress among adults** in 2023 BRFSS data. Model-based estimates are statistically smoothed values derived from small-area estimation models. Age-adjusted prevalence represents rates standardized to a common age distribution so that regions can be compared fairly.

Location	https://data.cdc.gov/500-Cities-Places/PLACES-County-Data-GIS-Friendly-Format-2025-release/i46a-9kgh/about_data
Format	CSV, JSON
Important variables	MHLTH_AdjPrev
Size	3,143 rows and 167 variables

Secondary Dataset: American Community Survey 5-Year Data (2009-2024)

This dataset provides demographic, socioeconomic, and housing information for communities across the United States, including county-level characteristics such as income, education, and employment. It includes Detailed Tables, Subject Tables, Data Profiles, and Comparison Profiles. We use **Data Profiles**, which summarize key indicators. Specifically, we use the **PE** fields. They report percentage estimates, which are statistical estimates derived from survey samples and adjusted to represent the population.

Location	https://www.census.gov/data/developers/data-sets/acs-5year.2024.html
Format	JSON
Important variables	Data Profile PE variables
Size	3,223 rows and 4,352 variables

Data Manipulation: Cleaning and Merging Datasets

Data Processing Environment

All data manipulation was performed in Python using Jupyter notebooks, providing a reproducible and auditable record of each transformation. All data were loaded into a pandas DataFrame for further processing.

PLACES - Key Column Creation

To merge the two datasets, we standardized the key identifiers. The PLACES dataset includes CountyFIPS as a numeric field, so we padded leading zeros to convert it into a five-character string as:

```
df['CountyFIPS'] =  
    df['CountyFIPS'].astype(str).str.zfill(5)
```

ACS - Loading and Combining

ACS data can be downloaded via API but we need to call 4 URLs for the following 4 Data Profile JSONs.

- DP02: Selected Social Characteristics in the United States
- DP03: Selected Economic Characteristics
- DP04: Selected Housing Characteristics
- DP05: ACS Demographic and Housing Characteristics

Each JSON is formed as an array, and its first line is variable names. When we load the data, we promote the first element as the column name. And to combine all 4 datasets, we created CountyFIPS column as a combination of the state and county columns as `df['CountyFIPS'] = df['state'] + df['county']` and merged the datasets as:

```
merged_df = df_dp2.merge(df_dp3, on='CountyFIPS',  
                        how='outer', suffixes=('_DP02', '_DP03'))  
merged_df = merged_df.merge(df_dp4, on='CountyFIPS',  
                        how='outer', suffixes=('', '_DP04'))  
merged_df = merged_df.merge(df_dp5, on='CountyFIPS',  
                        how='outer', suffixes=('', '_DP05'))
```

It is also used as key column to merge with PLACES dataset.

ACS - Loading Metadata

ACS Profile Data variable names are formatted as DP02_0001PE and they are not humanly understandable. To understand the meaning of each value, we load variable metadata (<https://api.census.gov/data/2024/acs/acs5/profile/variables.json>). And this information is also used to remove uncontrollable variables later.

Data Manipulation: Cleaning and Merging Datasets

ACS - Manually Clustering All Variables

Although the ACS Data Profile spans multiple domains, it does not provide predefined variable categories that suit our analysis. To reveal meaningful structure in the dataset, we clustered related variables prior to analysis based on the metadata, enabling us to identify key factors within each domain.

ACS - Removing Irregular Values

The ACS dataset includes special placeholder values, such as -666,666,666, which indicate errors or unavailable data. We replaced those irregular values with null values to ensure data integrity as:

```
for column in df.columns:
    if df[column].dtype in ['int64', 'float64']:
        df[column] = df[column].map( \
            lambda x: None if x < -200000000 else x)
```

ACS - Removing Empty Columns

Some ACS variables contain no valid observations for any county. Since such columns offer no analytical value, we removed them as

```
df = df.dropna(axis=1, how='all')
```

ACS - Removing Total Values

Our analysis uses ACS Data Profile PE (percentage estimate) values because percentage-based indicators are more suitable for regional comparison. But some PE fields contain total population counts and we removed them. Technically, we removed columns of int64 and ones including values more than 100.

ACS - Removing Uncontrollable Variables

The ACS dataset includes variables that represent human attributes, such as race, ancestry, sex, age, birth place, and language, which cannot be modified through regional policy or environmental interventions. We excluded these uncontrollable variables with variable metadata.

Merging PLACES and ACS datasets

We merged PLACES and ACS datasets with the column CountyFIPS. We used pandas merge method with inner combination as

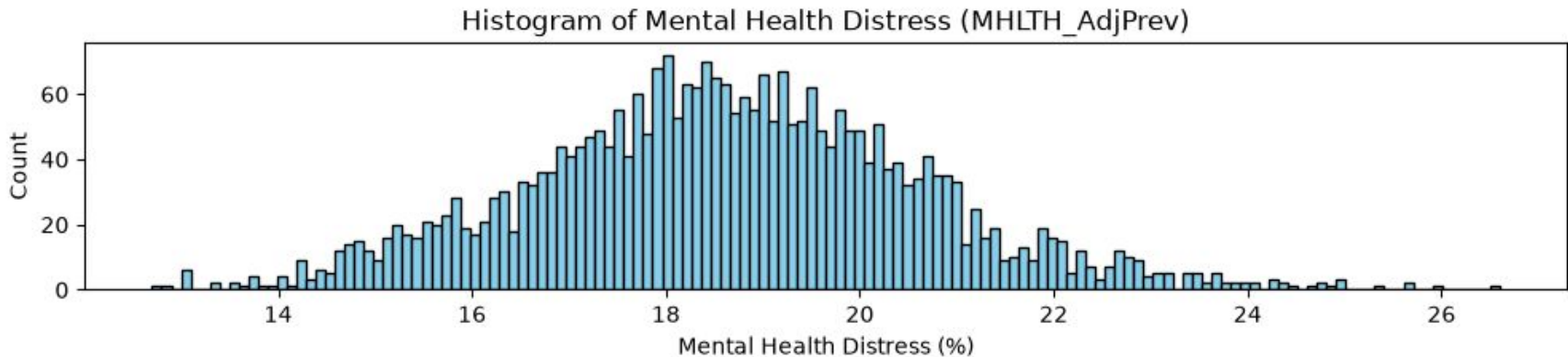
```
df = pd.merge(df_acs_profiles, df_places, on='CountyFIPS', how='inner')
```

Analysis & Visualization: Mental Health Distress Distribution

MHLTH_AdjPrev is the target variable in this analysis and represents the **age-adjusted prevalence of frequent mental distress** across U.S. counties. The distribution shows substantial county-level variation, with values ranging **from 12.7% to 26.6%** and **a median across counties of 18.6%**. **Most counties fall between 17% and 20%**, with some counties showing higher levels. This visualization provides descriptive context; the analysis focuses on associations between this indicator and regional characteristics, not on the distribution itself. The distribution is roughly bell-shaped, but the analysis does not rely on any assumption of normality.

Statistics of Mental Health Distress

count	2,956
mean	18.6
std	2.0
min	12.7
25%	17.3
50%	18.6
75%	19.9
max	26.6



Analysis & Visualization: Variable Selection and Correlation Analysis

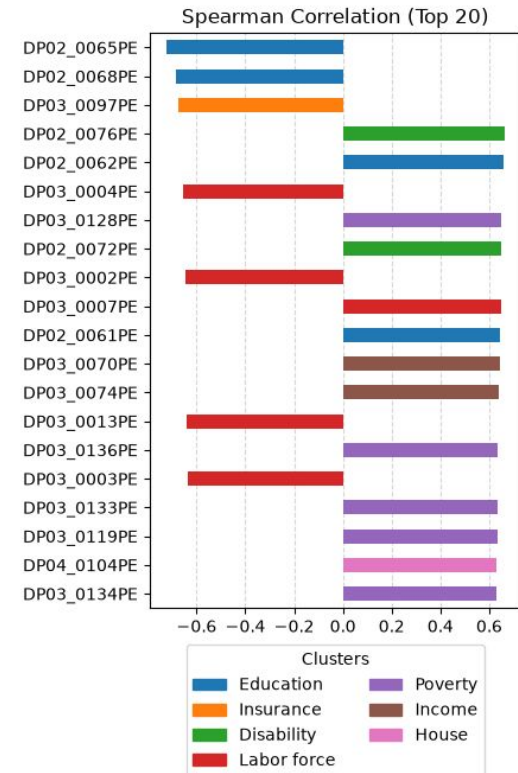
Selection of Representative Regional Characteristics

After data cleaning and filtering, 282 ACS variables were retained for correlation analysis. Spearman correlations were calculated between these variables and age-adjusted frequent mental distress. Of these, 61 variables had an absolute correlation greater than 0.50.

During exploratory data analysis, ACS variables were manually grouped into conceptual clusters using descriptions based on the Census variable metadata. From the highly correlated variables, one representative was selected from each of five distinct clusters. This approach reduced redundancy while retaining characteristics from different socioeconomic and health-related domains.

Spearman Correlation Analysis

Spearman correlation was used to examine the strength and direction of the associations between the selected regional characteristics and frequent mental distress across U.S. counties. **Spearman correlation** measures **monotonic relationships** without requiring a linear relationship or normally distributed variables and is **less sensitive to extreme values** than Pearson correlation. This makes it appropriate for the county-level ACS percentage estimates used in this analysis. The correlation results identified five regional characteristics with strong associations with frequent mental distress: educational attainment, employment, poverty, disability, and private health insurance coverage.

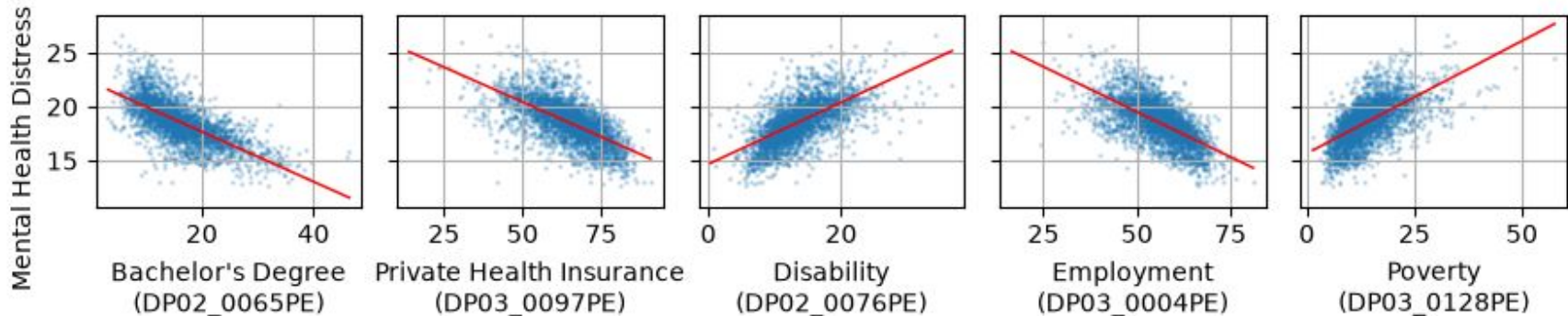


Analysis & Visualization: Key Variables Contribution for Mental Health

Among five identified key characteristics, **educational attainment (bachelor's degree)** showed the strongest negative association with frequent mental health distress with a coefficient of -0.72, followed by **insurance (private health insurance)** and **employment (employed)**. Conversely, **disability (with a disability)** and **poverty (below poverty level)** showed strong positive associations. It indicates that socioeconomic disadvantages align closely with regional mental health patterns.

Negative correlations indicate that counties with higher values tended to have lower frequent mental distress, while positive correlations indicate the opposite. These county-level associations are statistical correlations. They don't imply causal relationships.

Cluster	Variable	Spearman (ρ)	p-value
Education	DP02_0065PE (Bachelor's Degree)	-0.72	< 0.001
Insurance	DP03_0097PE (Private Health Insurance)	-0.67	< 0.001
Disability	DP02_0076PE (With a Disability)	0.66	< 0.001
Employment	DP03_0004PE (Employed)	-0.65	< 0.001
Poverty	DP03_0128PE (Below Poverty Level)	0.64	< 0.001



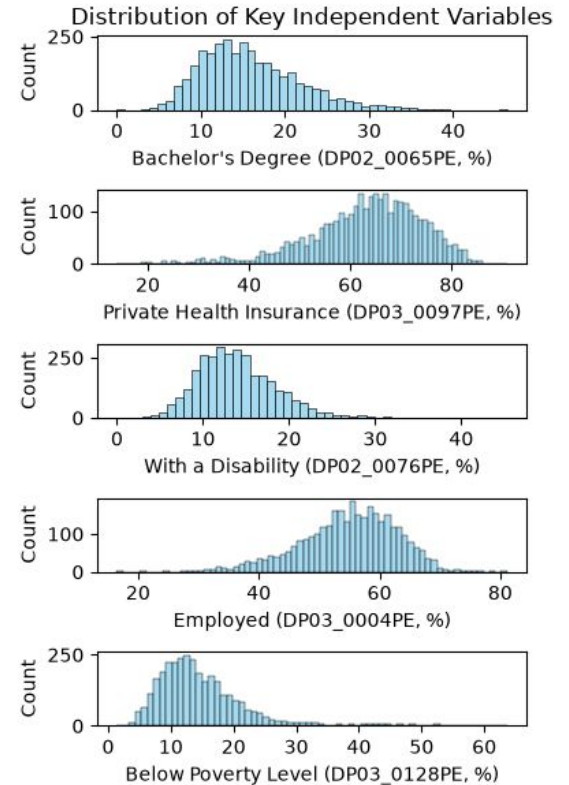
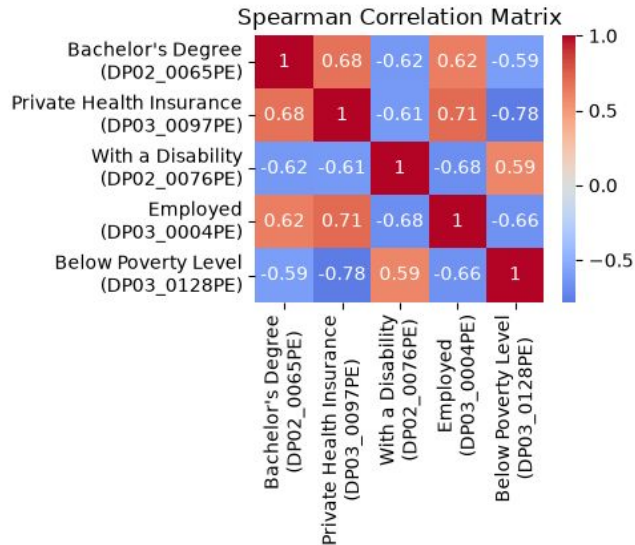
Analysis & Visualization: Key Variable Distributions and Interactions

These visualizations provide supporting context for interpreting the correlation results.

The histograms show how each of the five characteristics is distributed across U.S. counties. It reveals **differences in variability and skewness** that help contextualize their correlation strength.

The correlation heatmap illustrates how these characteristics relate to one another, highlighting **shared socioeconomic patterns**, such as the strong negative correlation ($\rho = -0.78$) between private health insurance and below poverty level.

These insights clarify the **structural associations** among the key regional characteristics. Also, they complement the main analysis.



Summary: Key Findings and Interpretation

Synthesizing the correlation results, this slide highlights the primary regional characteristics associated with frequent mental distress and outlines key interpretive contexts.

Key Findings & Ranking

Building on the correlation results, the analysis identified **five key regional characteristics** strongly associated with age-adjusted frequent mental distress across U.S. counties. Among these, **bachelor's degree has the strongest absolute correlation ($p = -0.72$)**, followed by private health insurance, disability, employment, and poverty statuses.

Insights by Domain

Negative Associations (Protective / Resource Factors): Educational attainment (bachelor's degree), private health insurance coverage, and employment rates show strong negative correlations. Counties with higher values in these indicators tend to exhibit lower frequent mental health distress.

Positive Associations (Higher Vulnerability / Disadvantage): Disability prevalence and concentrated poverty show strong positive correlations. Counties with higher rates in these indicators tend to exhibit greater frequent mental health distress.

Interpretation & Context

Overlapping Conditions: Educational, economic, and health-related conditions overlap significantly across counties. Consequently, these individual correlations **should not be interpreted as independent effects**.

Exploratory Patterns: These findings highlight broader regional patterns and are **exploratory in nature**, identifying geographic differences **rather than establishing direct causation**.

Limitations

Exploratory Variable Selection

Our variable selection process was exploratory. Manually clustering and filtering from a large pool of ACS indicators introduces **potential subjective choices** in feature selection.

Constraints of County-Level Spearman Correlation Analysis

The correlations represent county-level associations and **do not imply causation**. Although Spearman correlation reduces the influence of outliers by using rank data, it does not account for potential confounding among variables.

Uncertainty due to Estimation

Both ACS and PLACES are survey-based or model-based **estimations**. They introduce **uncertainty** especially for small counties, and potentially affect the strength of the correlations.

Monotonic Relationships

Spearman correlation captures only **monotonic relationships**. More complex or non-monotonic patterns between regional characteristics and mental-health outcomes cannot be detected. For those relationships, additional and complex modeling is required.

Ethical Considerations

Avoiding Misinterpretation of Aggregated Data

The results come from county-level, aggregated data. They **should not be used to make claims about individual people** in the counties. Using regional patterns to describe individuals can lead to fallacy. It may also unintentionally stigmatize communities that show higher levels of socioeconomic disadvantage or mental-health difficulties.

Responsible Use of Public Data

The analysis relies on publicly available data that does not include personal identifiers. Even so, careful interpretation is important, **especially for small counties where modeled estimates may be more sensitive to local conditions**. Counties should not be described as problematic or deficient.

Handling Correlations Carefully

This analysis identifies associations, not causes. **The results should not be used to justify specific interventions without additional research**. These correlations may help inform future studies or policy discussions, but they are not meant to recommend actions to particular communities.

Statement of Work

Work on this project was highly collaborative. We had utilized asynchronous communications among the team members. Each member of the team contributed substantially in brainstorming, retrieving, cleaning, analyzing, visualizing, and discussing data, as well as writing up code annotations and the report. The following is a more specific breakdown of responsibilities. Moving forward, incorporating more synchronous, joint meeting sessions will further enhance collaboration.

Anne Cao	Kenji Otsuka	Saleh Nasr
• Problem formulation. • Reviewing the technical report.	• Problem formulation. • Creation and management of Git repository. • Collecting, cleaning, manipulating datasets. • Exploratory Data Analysis including correlation analysis. • Designing and writing the technical report.	• Designing and writing the technical report • Reviewing and interpreting the correlation analysis. • Analyzing and summarizing key findings. • Identifying and documenting study limitations.

References

- Compton, M. T., & Shim, R. S. (Eds.). (2015). *The social determinants of mental health*. American Psychiatric Publishing.
- Bolduc, M. L. F., Saberi, P., Neilands, T. B., Mercado, C. I., Johnson, S. B., Freggens, Z. R. F., Banks, D., Njai, R., & Bullard, K. M. (2025). Economic factors associated with county-level mental health – United States, 2019. *PLOS ONE*, 20(6), e0300939. <https://doi.org/10.1371/journal.pone.0300939>
- Negaro, S. N. D., Hantman, R. M., Probst, J. C., Crouch, E. L., Odahowski, C. L., Andrews, C. M., & Hung, P. (2023). Geographic variations in driving time to US mental health care, digital access to technology, and household crowdedness. *Health Affairs Scholar*, 1(6), qxad070. <https://doi.org/10.1093/haschl/qxad070>